

Constraint Residual Detection: A Geometry-Based Approach to LLM Hallucination Detection

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Abstract

Large language models (LLMs) produce hallucinations with high confidence, rendering uncertainty-based detection ineffective on a substantial fraction of factual errors. On a 30-question false-premise subset of TruthfulQA, Qwen2.5-7B-Instruct produces confident wrong answers on 36.7% of questions—precisely the regime where predictive entropy and self-consistency methods fail ($\text{AUC} \leq 0.465$). We propose **Constraint Residual Detection**, a method that measures violations of the model’s internal constraint field $\Pi = \sum_i \nabla \sigma_i$ across the input-to-output transition. Rather than observing output confidence, we extract the constraint gradient field from hidden states and compute the norm change $\Delta \|\Pi\|$ as a hallucination signal. On TruthfulQA with Qwen2.5-7B-Instruct, our method achieves $\text{AUC} = 0.816$ (Cohen’s $d = 1.06$, $p = 0.018$), compared to 0.465 for predictive entropy and 0.250 for SelfCheckGPT. Cross-scale analysis reveals that the signal weakens substantially at smaller model sizes: $\text{AUC} = 0.532$ at 1.5B parameters (hidden dim 1536) versus 0.816 at 7B (hidden dim 3584). The method adds $\sim 28.5\text{ms}$ of computational overhead beyond standard generation (total latency $\sim 1.1\text{s}$).

1 Introduction

Hallucination—the generation of plausible but factually incorrect content—remains a critical barrier to deploying LLMs in production. The dominant detection paradigm measures output uncertainty: predictive entropy [2], token probability, or multi-sample consistency [3, 4]. These methods share a core assumption: incorrect outputs exhibit higher uncertainty than correct ones.

This assumption fails systematically on false-premise questions. When asked “Can sharks get cancer?”, a model confidently responds “No, sharks are immune to cancer”—and uncertainty-based detectors report low risk. The model *believes* its hallucination, so surface confidence decouples from correctness. On our 30-question false-premise evaluation set (TruthfulQA [1]), Qwen2.5-7B-Instruct hallucinates on 36.7% of questions (11/30), and competing detectors perform near or below random ($\text{AUC} \leq 0.465$).

We propose a different approach. Every LLM acquires implicit constraints during training—facts, logical relationships, syntactic structures. Content that violates these internalized constraints produces a detectable geometric signal in hidden state dynamics, *regardless of surface confidence*. This connects to recent work showing that truth-relevant information is linearly accessible in LLM representations [5, 6], but differs in measuring *constraint violation dynamics* across the input-to-output transition rather than probing static hidden states.

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Mathematically, we define the **constraint field** $\Pi(p) = \sum_i \nabla \sigma_i(p)$ where each σ_i measures a distinct dimension of internal constraint, and $\nabla \sigma_i$ captures token-level violation. The core empirical finding is:

$$\Delta \|\Pi\| = \|\Pi_{\text{output}}\| - \|\Pi_{\text{input}}\| \begin{cases} > 0 & (\text{hallucination: constraint violation increases during generation}) \\ \leq 0 & (\text{factual: constraint violation is stable or resolves}) \end{cases} \quad (1)$$

When the model hallucinates, it generates content that violates its internal constraints more than the input does, producing a positive $\Delta \|\Pi\|$. When it answers factually, constraint violation stays stable or decreases, yielding near-zero or negative $\Delta \|\Pi\|$. The detection signal is designed to be independent of surface confidence because it observes internal dynamics rather than output distributions.

2 Method

2.1 Constraint Functions

We operationalize five constraint functions $\sigma_i : \mathcal{H} \rightarrow [0, 1]$, each capturing a distinct dimension of internal model behavior. All σ_i are normalized to $[0, 1]$ with NaN/Inf guards.

Factual Consistency (σ_{fact}). We calibrate a truth direction $\mathbf{v}_{\text{truth}} \in \mathbb{R}^d$ from 10 contrastive prompt pairs (truthful vs. false completions), following evidence that truthfulness has a linear direction in activation space [5]. The direction is the normalized difference between mean truthful and mean false hidden states from the final transformer layer. Then:

$$\sigma_{\text{fact}}(h_t) = \frac{\cos(h_t, \mathbf{v}_{\text{truth}}) + 1}{2} \quad (2)$$

where $h_t \in \mathbb{R}^d$ is the hidden state at token t . Without calibration, two fallback proxies are available: when attention weights are accessible, the maximum attention value at position t serves as a decision-certainty proxy; otherwise, an L2-norm stability proxy is used: $\sigma_{\text{fact}}(h_t) = 1.0 - \|\|h_t\| - \bar{n}\|/\bar{n}$, where $\bar{n}$ is the empirical mean hidden state norm for the target model.

Syntactic Well-formedness (σ_{syntax}). Using attention weights $A_t \in \mathbb{R}^n$ at position t (averaged across heads):

$$\sigma_{\text{syntax}}(A_t) = 1 - \frac{H(A_t)}{H_{\text{max}}}, \quad H(A_t) = - \sum_{j=1}^n a_j \log a_j, \quad H_{\text{max}} = \log n \quad (3)$$

Low entropy (focused attention) indicates stronger syntactic constraint satisfaction.

Stylistic Coherence (σ_{style}). Given layer-wise hidden states $\{h_t^{(l)}\}_{l=1}^L$ at token t , we measure cross-layer norm stability via the coefficient of variation:

$$\sigma_{\text{style}}(\{h_t^{(l)}\}) = \exp \left(-3 \cdot \frac{\text{std}(\{\|h_t^{(l)}\|\})}{\text{mean}(\{\|h_t^{(l)}\|\})} \right) \quad (4)$$

Erratic norm variation across layers indicates register shifts or tonal inconsistency. The decay factor 3 was set on a small validation set.

Safety Alignment (σ_{safety}). Analogous to σ_{fact} , a refusal direction $\mathbf{v}_{\text{refusal}} \in \mathbb{R}^d$ is calibrated from 10 contrastive prompt pairs (refusal vs. compliant responses):

$$\sigma_{\text{safety}}(h_t) = \frac{\cos(h_t, \mathbf{v}_{\text{refusal}}) + 1}{2} \quad (5)$$

Without calibration, σ_{safety} defaults to 0.3 (neutral). This constraint captures the model’s safety-aligned operating regime, reflecting behaviors instilled during RLHF training [8].

Semantic Coherence ($\sigma_{\text{coherence}}$). Adjacent-token cosine similarity:

$$\sigma_{\text{coherence}}(h_{t-1}, h_t) = \frac{\cos(h_{t-1}, h_t) + 1}{2} \quad (6)$$

Sharp drops indicate semantic discontinuities that may signal confabulation.

2.2 Constraint Gradient Field

For each token position $t \geq 1$, per-constraint gradients are computed via finite differences:

$$\nabla \sigma_i(t) = \sigma_i(t) - \sigma_i(t-1) \quad (7)$$

yielding a 5-dimensional gradient vector $\nabla \sigma(t) \in \mathbb{R}^5$.

The constraint residual at token t is the signed sum:

$$\Pi(t) = \sum_{i=1}^5 \nabla \sigma_i(t) \quad (8)$$

Its magnitude $\|\Pi(t)\|$ captures the net constraint violation. We also define the cancellation ratio $c(t) = \|\sum \nabla \sigma_i\| / \sum \|\nabla \sigma_i\|$, which measures gradient component alignment: low $c(t)$ indicates opposing constraints, a signature of internal inconsistency. As a standalone detector, $c(t)$ performs near random (AUC = 0.528 at 1.5B), so we use it only as an auxiliary diagnostic rather than a primary detection signal.

2.3 Detection Signal

The detection signal is the mean residual magnitude shift from input to output:

$$\Delta \|\Pi\| = \frac{1}{|T_{\text{out}}|} \sum_{t \in T_{\text{out}}} \|\Pi(t)\| - \frac{1}{|T_{\text{in}}|} \sum_{t \in T_{\text{in}}} \|\Pi(t)\| \quad (9)$$

where T_{in} are content tokens in the input (excluding system prompts and special tokens), and T_{out} are generated output tokens. For hallucinations: $\Delta \|\Pi\| > 0$ (constraint violation increases). For factual answers: $\Delta \|\Pi\| \leq 0$ (constraint violation is stable or resolves).

In practice, hidden states for T_{in} are obtained by encoding the input prompt in a separate forward pass (with `output_hidden_states=True`), and states for T_{out} are obtained by re-encoding the generated output text independently. This means input constraint states are computed without autoregressive generation context, and output constraint states are computed without the prompt context. While this departs from the ideal of measuring constraint dynamics during generation, it avoids the engineering complexity of instrumenting the generation loop and adds negligible cost (see §3.6).

2.4 Calibration

To map the raw $\Delta\|\Pi\|$ signal to an interpretable probability, we fit logistic regression on a calibration subset:

$$P(\text{hallucination} \mid \Delta\|\Pi\|) = \frac{1}{1 + e^{-(\alpha + \beta \cdot \Delta\|\Pi\|)}} \quad (10)$$

Calibrated parameters for Qwen2.5-7B-Instruct are $\alpha = 2.85$, $\beta = 70.0$, obtained from 15 questions with evaluation on the remaining 15 (2-fold cross-validation, pooled). Since logistic regression is monotonic, the AUC of calibrated probabilities is identical to the raw score AUC; calibration improves interpretability, not discrimination. Risk tiers are defined as:

- **Low:** $P < 0.32$ — high confidence in correctness
- **Medium:** $0.32 \leq P < 0.52$ — uncertain, human review recommended
- **High:** $0.52 \leq P < 0.72$ — likely hallucination
- **Critical:** $P \geq 0.72$ — almost certain hallucination

We emphasize that with $n = 30$, calibration metrics such as Expected Calibration Error (ECE) are not reliably estimable. The probability values should be treated as preliminary risk indicators rather than precise probabilities. Larger-sample calibration is needed for production-grade reliability.

3 Experiments

3.1 Setup

We evaluate on TruthfulQA [1], a benchmark of 817 questions probing common human misconceptions across 38 categories. We focus on the false-premise subset—questions containing false assumptions—which is the hardest case for uncertainty-based methods: the model’s internalized misconceptions produce confident wrong answers.

Model: Qwen2.5-7B-Instruct (7B parameters, hidden dimension $d = 3584$, 28 layers, 28 query heads with grouped query attention, 4 KV heads).

Evaluation: $n = 30$ false-premise questions (11 hallucinatory, 19 correct; hallucination rate 36.7%). All experiments run on a single NVIDIA GPU with 24GB VRAM. Hidden states are extracted via `output_hidden_states=True` on separate encoding passes (input and output), rather than during the autoregressive generation step.

Baselines:

- **Predictive Entropy:** Token-level entropy of the output distribution [2].
- **Max Probability:** Maximum token probability as confidence proxy.
- **SelfCheckGPT** ($3\times$ sampling): Multi-sample consistency via pairwise BERTScore [3].
- **Response Length:** Heuristic based on response token count distributions.

All methods share the same underlying model inference for fair latency comparison.

Table 1: TruthfulQA Hallucination Detection Benchmark (Qwen2.5-7B-Instruct, false-premise subset, $n = 30$)

Method	AUC	Precision	Recall	Latency (ms)
Constraint Residual (Ours)	0.816	0.500	1.000	1,076
Predictive Entropy	0.465	0.167	0.333	1,005
Max Probability	0.447	0.167	0.333	1,006
SelfCheckGPT ($3\times$)	0.250	0.000	0.000	3,011
Response Length	0.443	0.250	0.500	1,004
<i>Cohen’s d (hallucination vs. correct separation)</i>	<i>1.06</i>	—	—	—
<i>p-value (Welch’s t-test, two-tailed)</i>	<i>0.018</i>	—	—	—

3.2 Main Results

Table 1 presents the results. Our method achieves $\text{AUC} = 0.816$, a $+0.351$ advantage over the best baseline (predictive entropy, $\text{AUC} = 0.465$). Our method achieves perfect recall (1.000) at 0.500 precision—detecting every hallucination while flagging half of the correct answers for review. This operating point was determined by optimizing the decision threshold on the evaluation set; out-of-sample recall may be lower. Nonetheless, the score distributions are well-separated (Cohen’s $d = 1.06$, $p = 0.018$), confirming that the underlying signal is robust.

The baselines collectively fail to discriminate. Predictive entropy and max probability hover near random ($\text{AUC} \approx 0.45\text{--}0.47$). SelfCheckGPT ($\text{AUC} = 0.250$) is *below* random: it detects zero hallucinations (precision = recall = 0.000), assigning “safe” to all samples.

3.3 Why Baselines Fail

The SelfCheckGPT result ($\text{AUC} = 0.250$, precision = recall = 0) reveals a structural failure mode. On false-premise questions, the model has internalized the misconceptions being probed. Sampling three times yields three *consistently wrong* answers, which the consistency check interprets as high agreement $\rightarrow$ “safe.” The method never flags a hallucination because consistency and correctness are inversely correlated on this task.

Predictive entropy and max probability fail because the model is well-calibrated to its own (incorrect) beliefs: it assigns high probability to wrong answers it has internalized as facts. These are not epistemic errors (the model is uncertain) but ontological errors (the model’s knowledge is wrong). Uncertainty metrics cannot detect what the model does not doubt.

3.4 Cross-Scale Analysis

We compare detection performance across two model scales.

Table 2: Detection Performance Across Model Scales				
Model	Hidden Dim	AUC	Cohen’s d	p -value
Qwen2.5-1.5B-Instruct	1536	0.532	-0.04	0.92
Qwen2.5-7B-Instruct	3584	0.816	1.06	0.018

At 1.5B scale ($d = 1536$), the signal is indistinguishable from random ($\text{AUC} = 0.532$, $p = 0.92$). Notably, Cohen’s d is negative (-0.04), indicating the signal direction reverses: at 1.5B, halluci-

natory outputs have slightly *lower* mean residual magnitude than correct ones, opposite to the 7B pattern. At 7B ($d = 3584$), performance rises to $\text{AUC} = 0.816$. This pattern is consistent with observations that factual knowledge representations strengthen at larger scales [7]. However, with only two model sizes evaluated, we cannot claim a phase transition; these results demonstrate a scale *difference* requiring validation on intermediate architectures (3B, 13B, 70B). The two models hallucinated on different subsets of the 30 questions (12 vs. 11 hallucinations for 1.5B and 7B, respectively), which is expected since model scale affects knowledge accuracy on individual questions.

3.5 Per-Constraint Ablation

We decompose the constraint residual at 1.5B scale, where per-token constraint states are available for all five dimensions.

Table 3: Per-Constraint Detection Performance (Qwen2.5-1.5B-Instruct, $n = 30$)

Constraint	AUC	Cohen’s d
$\Delta\sigma_{\text{fact}}$	0.634	0.30
$\Delta\sigma_{\text{style}}$	0.588	0.10
$\Delta\sigma_{\text{syntax}}$	0.472	−0.04
$\Delta\sigma_{\text{safety}}$	0.458	−0.10
$\Delta\sigma_{\text{coherence}}$	0.431	−0.14
$\Delta\ \Pi\ $ (Combined, 1.5B)	0.532	−0.04

At 1.5B, the factual consistency constraint ($\Delta\sigma_{\text{fact}}$, $\text{AUC} = 0.634$) is the strongest individual predictor, notably *outperforming* the combined 1.5B constraint residual ($\text{AUC} = 0.532$). This suggests that at smaller scales, gradient aggregation across constraints introduces noise that masks the factual signal. The stylistic coherence constraint ($\Delta\sigma_{\text{style}}$, $\text{AUC} = 0.588$) provides a moderate secondary signal.

At 7B, the combined constraint field reaches $\text{AUC} = 0.816$ —a substantial improvement over the 1.5B combined result (0.532) and exceeding any individual constraint measured at 1.5B, supporting the hypothesis that larger models develop richer constraint representations where multi-constraint aggregation becomes beneficial. Per-constraint decomposition at 7B scale requires storing all intermediate constraint states during generation and is deferred to future work.

3.6 Detection Latency

Our method uses three model passes: (1) an input encoding pass to extract hidden states from the prompt ($\sim 18.5\text{ms}$, using `output_hidden_states=True`), (2) the standard autoregressive generation pass ($\sim 1,047\text{ms}$), and (3) an output re-encoding pass to extract hidden states from the generated text alone ($\sim 10\text{ms}$), plus constraint computation ($\sim 1\text{ms}$). Total additional overhead beyond standard generation is $\sim 28.5\text{ms}$ (2.7% of the 1,076ms total). This is comparable to single-inference baselines (predictive entropy: 1,005ms, which requires only logits from the generation pass) and substantially faster than multi-sample approaches (SelfCheckGPT: 3,011ms for $3\times$ sampling). The input and output encoding passes operate on short sequences (< 256 tokens) and add negligible cost relative to autoregressive generation.

4 Discussion

4.1 Comparison to Related Work

Uncertainty-based methods [2, 4] estimate output confidence. They fail when the model is confident in its errors—precisely the regime of false-premise questions.

Consistency-based methods [3] compare multiple outputs, assuming errors vary while truths are stable. On false-premise questions, this inverts: correct outputs vary (the model reasons through a false premise) while incorrect outputs are consistent (the model reliably recites its misconception).

Probe-based methods [5, 6] train classifiers on hidden states. Our approach shares the intuition that hidden states encode truth-relevant information, but uses *dynamic constraint gradients* across the generation process rather than static probes. This captures the process of constraint violation as it unfolds.

Commercial hallucination detection (Galileo, Arthur, NeMo Guardrails) relies on uncertainty or rules. Constraint residual detection is complementary: it captures a geometric signal invisible to surface-level methods.

4.2 Limitations

Sample size: Evaluation used $n = 30$ false-premise questions (11 positive, 19 negative). While the effect size is large (Cohen’s $d = 1.06$, $p = 0.018$), the small sample means AUC variance is itself uncertain. Full-benchmark evaluation on all 817 TruthfulQA questions, with proper confidence intervals via bootstrap, is needed to establish reliable performance bounds.

Calibration scale: Logistic calibration on 30 samples yields preliminary α, β estimates. The calibration code is currently hardcoded from the fitted values; the exact cross-validation procedure is not independently scripted, limiting reproducibility. Expected Calibration Error (ECE) is not reliably estimable at this sample size. Production deployment should use larger, multi-domain calibration sets with versioned calibration scripts.

Threshold tuning: The precision and recall reported in Table 1 reflect a decision threshold optimized on the evaluation set. Out-of-sample precision and recall at the same threshold will likely be lower. The AUC metric (0.816), being threshold-independent, is the more reliable performance indicator.

Model specificity: Calibration parameters (α, β) and truth/refusal directions are specific to Qwen2.5-7B-Instruct. Cross-model generalization has not been evaluated.

Language: English TruthfulQA only. Cross-lingual performance is unexplored.

Latency: ~ 1.1 s is suitable for post-hoc screening but not for interactive-speed streaming.

Scale coverage: Two model sizes (1.5B, 7B). Intermediate scales and other architectures (Llama, Mistral) are needed to characterize scaling behavior.

4.3 Future Work

- **Full-benchmark evaluation:** TruthfulQA 817 questions with bootstrap confidence intervals.
- **Multi-model calibration:** Llama-3, Mistral, DeepSeek architectures; cross-model transfer of constraint directions.
- **Token-level streaming:** Real-time $\Pi(t)$ computation during autoregressive generation for early interception.

- **Cross-lingual validation:** Chinese TruthfulQA, multilingual MMLU.
- **RAG integration:** Compound detection combining constraint residual signals with retrieval grounding.
- **Scaling characterization:** Intermediate scales (3B, 8B, 13B, 70B) to map the constraint field’s scaling behavior.

5 Conclusion

We introduced Constraint Residual Detection, a method that detects hallucinations by measuring violations of a model’s internal constraint field rather than output uncertainty. On TruthfulQA false-premise questions with Qwen2.5-7B-Instruct, our method achieves $AUC = 0.816$ with perfect recall (1.000), substantially outperforming uncertainty-based baselines (best competitor $AUC = 0.465$, Cohen’s $d = 1.06$, $p = 0.018$). At 1.5B scale, the signal degrades to near-random ($AUC = 0.532$), with only the factual consistency constraint retaining discriminatory power ($AUC = 0.634$), suggesting that multi-constraint aggregation requires sufficient representational capacity.

The key insight is that LLMs *believe* their hallucinations on false-premise tasks—surface confidence is orthogonal to correctness. By observing internal constraint dynamics rather than output distributions, we detect the geometric signature of fabrication that uncertainty-based methods are blind to. The method adds modest computational overhead (~ 28.5 ms beyond standard generation) and produces monotonic probability estimates via logistic regression, making it practical for hallucination screening when integrated with the generation pipeline.

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